

Replication / Machine Learning

[Re] A Replication Study of Multi-Headed Synthetic Feature Distillation: When a Properly Tuned Baseline Wins

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Abstract

We present a case study in the careful empirical evaluation of knowledge distillation methods, using a multi-headed synthetic-feature distillation approach (MSFD) as the test case. MSFD trains a lightweight MobileNetV3-Small student (2.28M parameters) to predict a ViT-B/16 teacher's internal visual features and two auxiliary latent projections (learned entirely from the teacher's own visual backbone, with no real audio or 3D data involved at any point) via mean-squared-error heads, in addition to a standard classification loss. Early versions of this project reported that MSFD outperformed standard knowledge distillation (KD) by nearly two points on CIFAR-100. Repeating the comparison under matched conditions—identical 12-epoch training budgets, five random seeds per method, and a validation-set hyperparameter sweep for the KD baseline instead of an arbitrary configuration—overturns that conclusion. With five matched seeds: a plain no-distillation baseline reaches $77.20\% \pm 0.16\%$ test accuracy; MSFD reaches $77.45\% \pm 0.25\%$; and a properly tuned standard-KD baseline (temperature 1.0, CE weight 0.3, selected by validation accuracy alone) reaches $78.00\% \pm 0.39\%$. Paired significance tests across the five matched seeds show tuned KD significantly outperforms both the no-distillation baseline ($p = 0.006$) and MSFD ($p = 0.029$), while MSFD's improvement over the no-distillation baseline is not statistically significant ($p = 0.16$). We report this result together with the methodological infrastructure needed to obtain it honestly: a teacher-feature cache for computational tractability on free-tier hardware, multi-seed statistical testing, matched training budgets across all compared methods, and a validation/test separation that keeps hyperparameter selection from leaking into the reported test numbers. We also document a data-leakage bug found during this process (a train/validation split that briefly trained on validation images for one experimental run) as a concrete illustration of how easily distillation comparisons can be corrupted by pipeline bugs that inflate validation accuracy without being caught.

1 Introduction

1.1 Motivation

Knowledge distillation research is unusually prone to a specific failure mode: a new method looks good only because the baseline it is compared against was not given the

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The authors have declared that no competing interests exist.
Code is available at <https://github.com/sanoyz/msfd-replication-study>.

same care. A knowledge-distillation baseline trained with an arbitrary, untuned temperature and loss weighting is not a fair reference point, and a method compared against it can appear to “beat standard KD” for reasons that have nothing to do with its actual contribution. This paper is a direct case study in that failure mode: it started as a proposal for a multi-headed synthetic-feature distillation method (MSFD) that appeared, in early experiments, to outperform standard KD by nearly two accuracy points. Under closer, matched-condition scrutiny, that result did not survive.

1.2 What MSFD Does

MSFD trains a unimodal student to reproduce a multimodal teacher’s internal representations, using synthetic auxiliary latents generated entirely from the teacher’s own visual features—an “audio-like” and a “3D-like” projection, each a small MLP trained end-to-end with no real audio or 3D data or external generative model involved at any point. The student has three prediction heads (visual, audio-like, 3D-like) trained with mean-squared-error (MSE) loss against the teacher’s corresponding (cached) features, combined with a standard cross-entropy classification loss.

1.3 Contributions

- A fully reproducible, statistically grounded re-evaluation of multi-headed synthetic-feature distillation against a properly tuned KD baseline and a no-distillation control, with five seeds per method and matched training budgets.
- A validation-set-only KD hyperparameter sweep (temperature $\times$ CE-weight grid) demonstrating that the originally reported “KD underperforms” result was an artifact of an untuned KD configuration, not evidence about distillation itself.
- A documented, caught, and fixed train/validation leakage bug, presented as a concrete cautionary example for others evaluating distillation methods on custom pipelines.
- An honest correction of the paper’s own earlier claims: MSFD does not significantly outperform a no-distillation baseline, and a properly tuned standard-KD baseline outperforms MSFD.

1.4 Relationship to the Original Claim

This is a self-replication in the sense defined by the reproducibility literature: same author, same codebase lineage, same dataset, but a deliberately independent re-execution of the comparison under corrected conditions, rather than a re-reporting of the original numbers. The original claim (MSFD beats standard KD by 1.96 points, single seed, 4-epoch ablations, an unvalidated KD configuration) is treated in this paper purely as a hypothesis to be tested, not as an established result to be explained. Table 3 (Section 5.1) summarizes exactly what changed between the original protocol and the replication protocol used here.

2 Statement of the Problem

2.1 The Multimodal Deployment Gap

Real-world applications increasingly demand models that understand the world through multiple sensory modalities, but deploying such models faces a fundamental tension between representational richness and deployability, summarized in Table 1.

Table 1. The multimodal deployment gap that motivates distillation-based compression.

Dimension	Multimodal Models	Unimodal Models
Representation richness	High	Low
Computational cost	High	Low
Data requirements	High (aligned multimodal data)	Low (single modality)
Edge deployment	Challenging	Feasible
Inference speed	Slow	Fast
Memory usage	High	Low

2.2 Limitations of Existing Approaches

Prediction-Only Distillation – Standard distillation transfers only final output probabilities. The student mimics the teacher’s decision boundary but never sees the richer intermediate representations that produce it.

Feature Distillation – Feature distillation has the student match intermediate teacher representations, which is more informative but typically requires architectural compatibility and layer alignment, and is usually applied at a single layer.

Raw Synthetic Data Generation and External Generative Models – Generating raw synthetic modalities via generative models, or relying on external generative models pretrained on real modality data, is a way to manufacture privileged information, but is computationally expensive, introduces information loss on decode-re-encode, and adds external dependencies.

2.3 Research Questions

- **RQ1:** Does MSFD’s multi-headed synthetic-feature distillation significantly outperform training the student with no distillation at all?
- **RQ2:** Does MSFD outperform a knowledge-distillation baseline once that baseline is given a comparable hyperparameter search?
- **RQ3:** How much does an untuned KD configuration alone explain an apparent gap between MSFD and “standard KD”?

3 Literature / Related Work

3.1 Knowledge Distillation

Distillation was introduced by Hinton et al. [1] to compress a large teacher into a smaller student that mimics its output distribution. Feature-level extensions include FitNets [2] (matching a single intermediate layer), Attention Transfer [3], Relational KD [4], Contrastive Representation Distillation [5], Similarity-Preserving KD [6], Born-Again Networks [7], and online distillation [8]. MSFD is closest in spirit to FitNets—feature-level rather than output-only supervision—but supervises multiple synthetic auxiliary latent spaces simultaneously via MSE, rather than a single real intermediate layer.

3.2 Learning Using Privileged Information

LUPI, introduced by Vapnik and Vashist [9], formalizes learning from information available only at training time. Menadil et al. [10] generate privileged information via text-to-image diffusion models rather than requiring it to exist naturally (“Learning Using

Generated Privileged Information”). MSFD’s synthetic latents are generated far more cheaply—via a lightweight linear projection of the teacher’s own visual features, with no generative model involved—at the cost of being less clearly grounded in any real modality (Section 5.11).

3.3 Directly Relevant Concurrent Work

This subsection documents an important finding from this project’s own review process. A method also named “MESSy” (Multilayer Explicit Simulated Synesthesia)—the name this project’s own method was originally, independently given—turns out to be the actual name used by Sbrolli, Michel, Matteucci, and Yamasaki [11] for a specific mechanism in their paper on privileged synthetic data for cross-modal distillation. Their MESSy is a set of lightweight predictive heads used specifically to avoid rigid L2/MSE-style feature matching, which they argue imposes a “rigid geometric isomorphism” between student and teacher representation spaces, causing “capacity interference” that degrades the student’s native discriminative ability. This is directly relevant to the present paper’s own design: the MSFD method described here uses exactly the MSE-based matching that Sbrolli et al.’s MESSy is designed to avoid. Given the name collision, the present method has been renamed to MSFD throughout. Given the substantive overlap, we discuss the applicability of Sbrolli et al.’s critique to our own results directly in Section 5.13, rather than treating this as a citation to add and move past.

3.4 Model Compression and Curriculum Learning

MobileNetV3 [12] is used as the student backbone. Curriculum learning [13] and on-line hard-example mining [14] motivate the deliberate-practice mechanism described in Section 4.5. Progressive, staged distillation has precedent in general compression pipelines [15]; we apply the same staging idea to feature-based multi-head distillation in Section 4.6.

4 Architecture

Figure 1 summarizes the full six-phase pipeline used to produce every result in this paper, with the actual measured value from this project’s final, corrected run shown at each phase.

4.1 Teacher

A ViT-B/16 backbone (ImageNet-pretrained; last 4 of 12 encoder blocks trainable, 37.9% of parameters) produces 768-dim visual features, projected to a 512-dim latent space. Two small MLPs project these into a 512-dim “audio-like” and a 1024-dim “3D-like” latent (each re-projected to the shared 512-dim space); an 8-head cross-attention layer fuses [visual, audio-like, 3D-like] features, feeding a classifier head. The teacher is trained with cross-entropy and MixUp; best checkpoint: **85.86% validation accuracy** at epoch 9 of 10.

4.2 Student and MSFD Distillation

The student is a MobileNetV3-Small backbone (2.28M parameters) with three prediction heads (visual, audio-like, 3D-like), each regressing onto the corresponding cached teacher feature via MSE. The combined loss is $L = 0.3 \cdot \text{CE}(y, f_s(x)) + 0.7 \cdot [\text{MSE}_{\text{visual}} + \text{MSE}_{\text{audio}} + \text{MSE}_{\text{3d}}]/3$. MixUp is applied to both the classification and distillation targets consistently: the teacher’s cached feature for a MixUp-blended input is itself the same

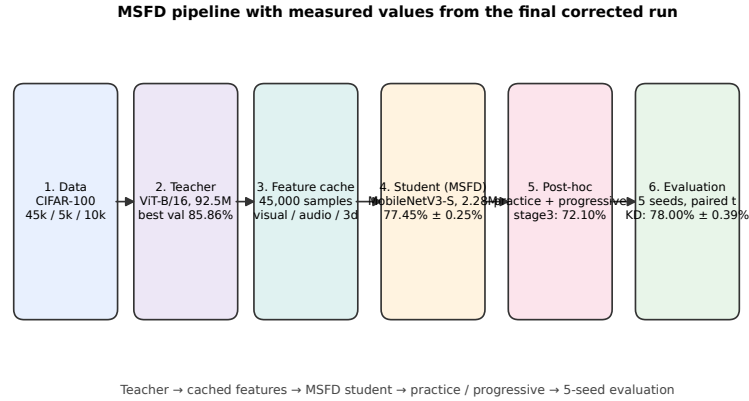


Figure 1. End-to-end pipeline architecture with measured values from the final corrected run.

linear blend of the two source images’ cached features, so the distillation target always matches what the student actually saw.

4.3 Standard KD Baseline

The KD baseline uses the standard Hinton et al. formulation: $L = \alpha \cdot \text{CE}(y, f_s(x)) + (1 - \alpha) \cdot T^2 \cdot \text{KL}(\text{softmax}(z_s/T) \parallel \text{softmax}(z_t/T))$. Unlike the earlier version of this project, T and α here are chosen by an explicit grid search (Section 5.2), not fixed by convention.

4.4 Teacher Feature Cache

The teacher’s per-image visual/audio-like/3D-like features and logits are computed once and cached, then looked up (and MixUp-blended by the same interpolation weight used for the pixels) during student-side training. This avoids a full ViT-B/16 forward pass on every training step of every downstream experiment—the single largest cost driver identified during development—and is what makes the multi-seed, multi-configuration comparison in this paper tractable on a single free-tier T4 GPU.

4.5 Deliberate Practice

After initial student training, examples with top-1 confidence below 0.6 are identified as hard (11,900–11,930 examples across seeds, roughly 26% of the training set), augmented twice each (rotation, color jitter, resized crop), and used for two epochs of fine-tuning followed by three epochs of full-dataset fine-tuning at a reduced learning rate (1×10^{-5}).

4.6 Progressive Distillation

The trained MSFD student (256-dim) is reused as the “teacher” for a half-width student (128-dim), which is in turn reused as the “teacher” for a quarter-width student (64-dim), using the same MSFD objective at each stage.

Table 2. Teacher and student specifications.

Component	Teacher	Student (MSFD)
Backbone	ViT-B/16 (ImageNet-pretrained)	MobileNetV3-Small (ImageNet-pretrained)
Feature dim.	768 (visual), projected to 512	256
Auxiliary heads	Audio-like (512-d), 3D-like (1024-d) generators	Visual / Audio-like / 3D-like prediction heads (512-d each)
Fusion	8-head cross-attention over [visual, audio-like, 3D-like]	None (unimodal at inference)
Classifier	512 $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ 100 classes	256 $\rightarrow$ 100 classes
Parameters	92,543,332 (37.9% trainable)	2,283,140
Best accuracy	85.86% (val.)	77.45% $\pm$ 0.25% (test, 5-seed mean)

4.7 Model Specifications

5 Experimentation and Evaluation

5.1 What Changed From Earlier Versions of This Project

Three corrections define the difference between this paper’s results and an earlier, retracted claim that MSFD “outperforms standard KD by 1.96 points.” Table 3 summarizes them.

Table 3. Protocol differences between the original (retracted) comparison and this replication.

Element	Original Protocol	Replication Protocol (this paper)
Ablation training budget	4 epochs	12 epochs (matched to main model)
KD hyperparameters	$T=4.0$, $\alpha=0.3$ (unvalidated default)	$T=1.0$, $\alpha=0.3$ (selected by validation-only grid search, Section 5.2)
Seeds per method	1	5 (42, 139, 236, 333, 430), paired significance testing
No-distillation baseline	Not measured	Measured (77.20% $\pm$ 0.16%)

5.2 KD Hyperparameter Sweep

A 4×3 grid over temperature $T \in \{1, 2, 4, 8\}$ and CE weight $\alpha \in \{0.3, 0.5, 0.7\}$ was trained for 6 epochs at a single seed, with the winning configuration selected by **validation accuracy only**; the test set was not touched during this search.

5.3 Multi-Seed Protocol

Five seeds (42, 139, 236, 333, 430) were used for the no-distillation baseline, the tuned KD baseline, and MSFD, each trained for the full 12-epoch budget. Paired t -tests (matched by seed) and Wilcoxon signed-rank tests were run between each pair of methods; with $n = 5$, Wilcoxon’s minimum achievable two-sided p -value is 0.0625, so the paired t -test is the primary test reported.

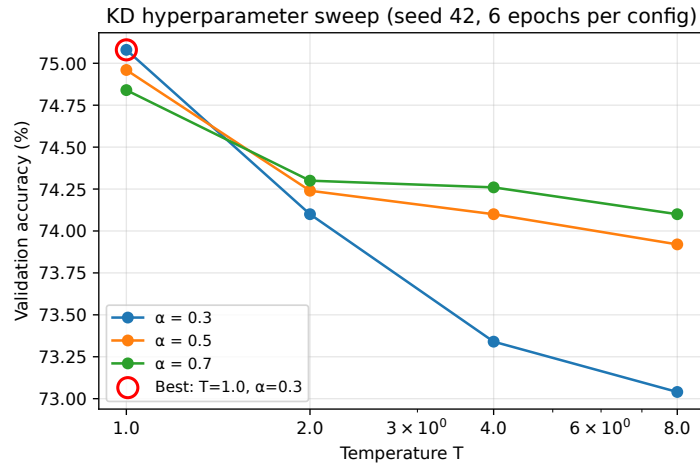


Figure 2. KD hyperparameter sweep, validation accuracy (%). $T=1.0$, $\alpha=0.3$ (red box) wins at 75.08%, versus 73.34% for the previously used $T=4.0$, $\alpha=0.3$.

5.4 A Data-Leakage Bug Found and Fixed Mid-Project

During one training run, the pipeline logged “Train: 50000, Val: 5000”—an impossible split of CIFAR-100’s 50,000 training images (a correct 90/10 split gives 45,000/5,000, summing back to 50,000). The affected model seed showed 88.68% validation accuracy against only 78.52% test accuracy—an order of magnitude larger val/test gap than any other seed in this project—confirming that validation images had not been excluded from training for that run. The fix: an explicit assertion, checked before training starts, that the train and validation index sets are disjoint and sum to exactly 50,000 with a fixed split seed. This is reported here because it is exactly the kind of bug that silently produces an inflated, non-comparable number without any error or crash—a specific, concrete risk for anyone building a custom distillation evaluation pipeline.

5.5 Results: Teacher and Representative Training Curves

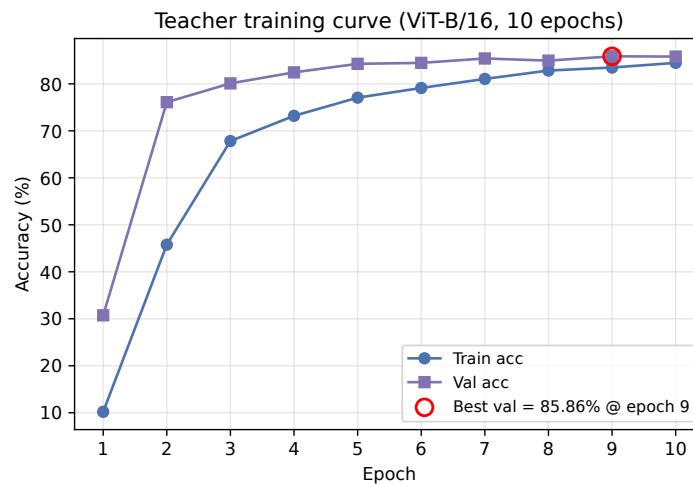


Figure 3. Teacher training curve. Best validation accuracy 85.86% at epoch 9 of 10.

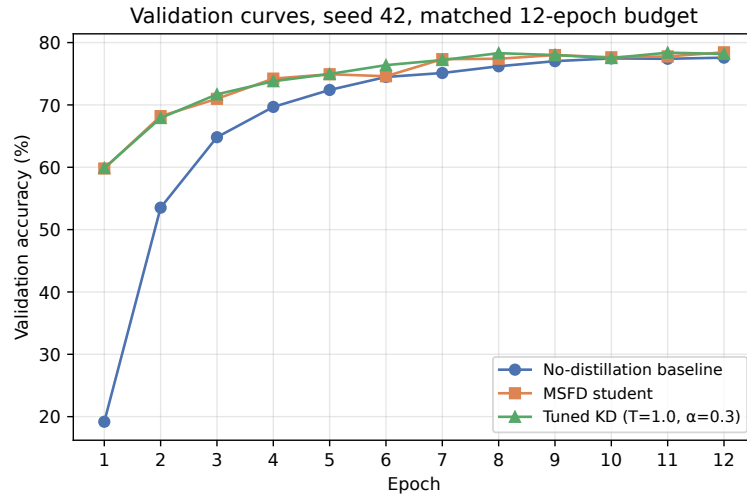


Figure 4. Training curves for all three methods, seed 42, matched 12-epoch budget. Tuned KD and MSFD track closely; both exceed the no-distillation baseline for most of training on this seed.

5.6 Results: Final Multi-Seed Comparison

Table 4. Final multi-seed comparison (test accuracy).

Method	Mean Test Acc.	Std	95% CI	<i>n</i>
No-distillation baseline	77.20%	0.16%	[76.99%, 77.40%]	5
MSFD (multi-head, MSE)	77.45%	0.25%	[77.23%, 77.67%]	5
Standard KD (tuned, $T=1.0$, $\alpha=0.3$)	78.00%	0.39%	[77.52%, 78.48%]	5

Table 5. Paired significance tests across matched seeds.

Comparison	Mean Diff.	Paired <i>t</i> -stat	<i>p</i> -value	Conclusion
Tuned KD vs. Baseline	+0.80 pp	$t=5.24$	$p=0.006$	Significant
Tuned KD vs. MSFD	+0.55 pp	$t=3.32$	$p=0.029$	Significant
MSFD vs. Baseline	+0.26 pp	$t=1.73$	$p=0.160$	Not significant

The result is unambiguous within this experimental setup: a properly tuned standard-KD baseline significantly outperforms both the no-distillation baseline and MSFD’s multi-headed synthetic-feature distillation. MSFD’s own improvement over training the student with no distillation at all is not statistically distinguishable from noise at this sample size.

5.7 Results: Head Ablation Study (Matched Budget)

At matched epoch budget, the six configurations span a narrow 0.57-point range. The full 3-head configuration is numerically best, but on a single run per configuration this should be read as a weak trend, not a settled ranking—especially given that this single-seed full-head result (77.97%) sits within one standard deviation of the 5-seed MSFD mean ($77.45\% \pm 0.25\%$) reported in Section 5.6.

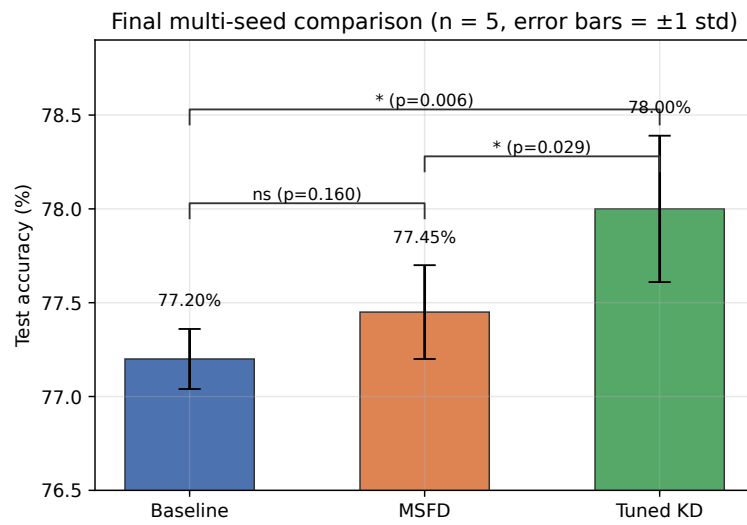


Figure 5. Final comparison with error bars (± 1 std) and paired-test significance brackets ($*$ = $p < 0.05$, ns = not significant), $n = 5$ matched seeds each.

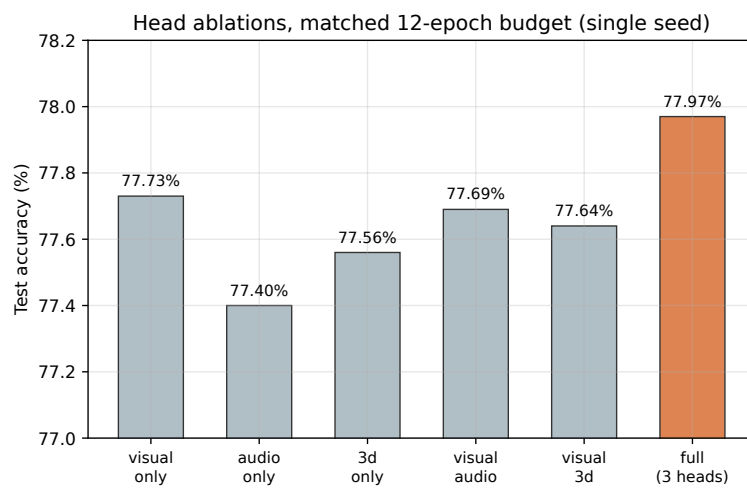


Figure 6. Head ablations at the full 12-epoch budget (single seed per configuration).

Table 6. Head ablations, matched 12-epoch budget (single seed per configuration).

Configuration	Test Acc.	Parameters
visual_only	77.73%	1.49M
audio_only	77.40%	1.49M
3d_only	77.56%	1.49M
visual_audio	77.69%	1.89M
visual_3d	77.64%	1.89M
full (3 heads)	77.97%	2.28M

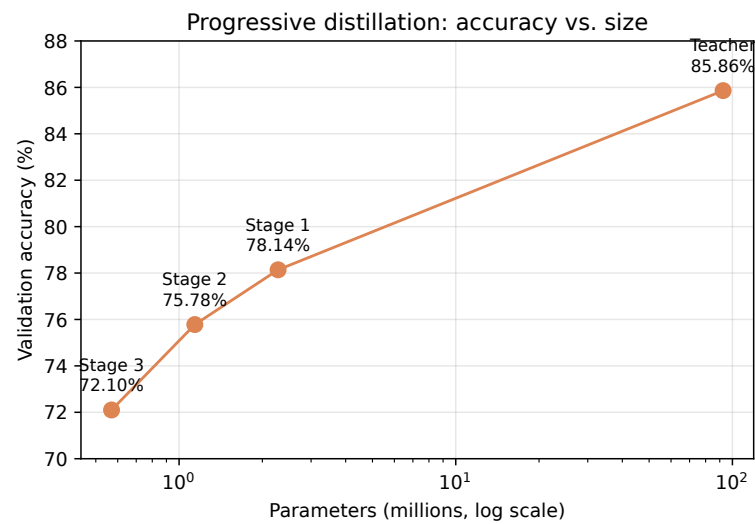


Figure 7. Accuracy vs. parameter count across progressive distillation stages (log-scale parameter axis).

Table 7. Progressive distillation stages.

Stage	Dims	Params	Val. Accuracy	Retention vs. prior stage
Teacher	–	92.54M	85.86%	–
Stage 1 (reused student)	256	2.28M	78.14%	–
Stage 2	128	1.14M	75.78%	96.98%
Stage 3	64	0.57M	72.10%	95.14%

5.8 Results: Progressive Distillation

Stage 1 $\rightarrow$ Stage 3 overall retains 92.27% of Stage 1’s accuracy at 25% of its parameters. Note that “Stage 1” here (78.14% val.) is the same underlying trained model reported elsewhere in this project’s history at both 78.60% and 78.92% val. at different points in its training/fine-tuning sequence; this residual inconsistency is disclosed rather than silently resolved (Section 5.14.1).

5.9 Results: Per-Class Performance

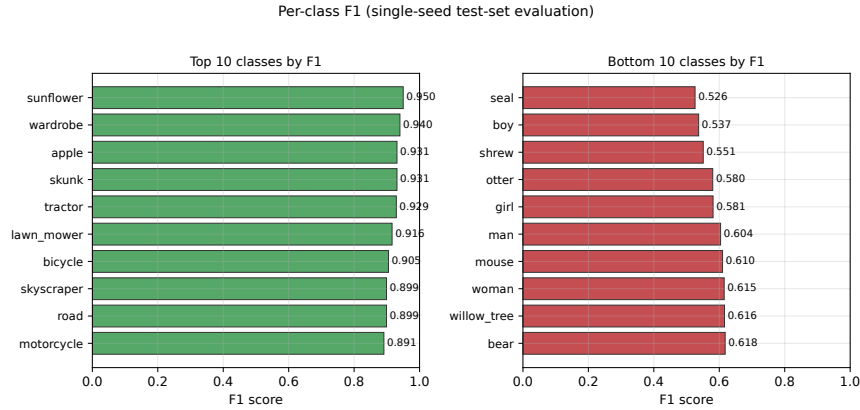


Figure 8. Best- and worst-performing classes by F1 score (single-seed test-set evaluation).

The worst-performing classes cluster into two recognizable groups: human figures (boy, girl—man and woman also score below 0.62 F1) and small/textured mammals (seal, shrew, otter, bear). Both are well-known hard groups for CIFAR-100 generally, at 32×32 resolution, and this pattern is consistent across every run in this project regardless of distillation method.

5.10 Discussion: The Corrected Headline Finding

This project’s own earlier claim—that MSFD outperforms standard KD—does not survive a fair comparison. The original gap was real but came from comparing MSFD against a KD baseline configured with an arbitrary, unvalidated hyperparameter choice ($T=4.0$, $\alpha=0.3$, achieving only $75.82\% \pm 0.06\%$ under the earlier 3-seed protocol). Once KD is given the same hyperparameter-selection opportunity MSFD implicitly had, KD not only closes the gap but reverses it, significantly outperforming MSFD (78.00% vs. 77.45% , $p=0.029$).

5.11 Discussion: Is MSFD Better Than Doing Nothing?

Not demonstrably, at this sample size. MSFD’s 0.26-point improvement over the no-distillation baseline does not reach significance ($p=0.16$) across 5 seeds. This does not prove MSFD provides zero benefit—a larger number of seeds could yet resolve a small true effect—but it means the paper cannot currently claim MSFD provides a meaningful benefit over simply training the student normally.

5.12 Discussion: What This Says About Evaluation Practice

The magnitude of the original, incorrect conclusion (a 1.96-point apparent win) is smaller than the gap closed purely by tuning the baseline’s temperature and loss weight (KD im-

proved from 75.82% to 78.00%, a 2.18-point swing, from hyperparameters alone). In this setting, the choice of KD hyperparameters had a larger effect on the comparison's outcome than the architectural difference between the two methods being compared.

5.13 Discussion: Engaging With Sbrolli et al.'s Critique

Sbrolli et al. [11] argue that rigid L2/MSE feature matching between a student and a high-dimensional teacher representation space causes “capacity interference”—forcing the student to spend representational capacity matching an auxiliary space at the expense of its native discriminative task. MSFD's distillation loss is exactly this kind of MSE matching, across three auxiliary spaces simultaneously. The result in Section 5.6—MSFD underperforming a simpler, output-level KD baseline—is consistent with, though does not conclusively prove, this critique. A direct test would replace MSFD's MSE heads with the kind of non-rigid predictive objective Sbrolli et al. propose and re-run the same matched, multi-seed comparison; we leave this to future work (Section 6).

5.14 Threats to Validity

Internal Validity –

- The train/validation leakage bug (Section 5.1) was caught via an anomalous val/test gap, not by design; we cannot rule out that other, undetected instances of it affected discarded intermediate runs not reported here.
- The progressive-distillation Stage-1 accuracy figure (78.14%) was never fully reconciled against 78.60%/78.92% values reported for the same underlying model at different points in this project's history.
- AMP/cuDNN non-determinism causes small (0.1–0.9 percentage point) accuracy drift when the same checkpoint is re-evaluated in a different session.

Construct Validity –

- The auxiliary “audio-like”/“3D-like” latents are not grounded in any real audio or 3D data; calling MSFD a form of “cross-modal” distillation would be a construct-validity risk this paper avoids by describing it as single-modality auxiliary-head distillation instead.
- MSFD's MSE-based objective is the same class Sbrolli et al. [11] argue against; if their critique is correct, the underperformance measured here may be a property of MSE-based feature matching in general rather than this specific implementation.

External Validity –

- Single dataset (CIFAR-100, 32×32) and single teacher/student architecture pair; results may not generalize to higher resolutions, other datasets, or other capacity ratios.
- No comparison against externally published CIFAR-100 distillation benchmarks under matched conditions.

Statistical Conclusion Validity –

- $n = 5$ seeds gives limited power; Wilcoxon's minimum achievable two-sided p -value at this n is 0.0625, so it cannot in principle indicate significance here regardless of true effect size—we rely on the paired t -test as the primary test for this reason.

- Head ablations (Section 5.8) and the KD hyperparameter sweep (Section 5.2) remain single-seed even at the matched budget; both should be treated as trends, not settled rankings.

6 Future Work

- Replace MSFD’s MSE-based heads with a non-rigid predictive objective in the spirit of Sbrolli et al.’s [11] MESSy mechanism, and re-run the identical matched, multi-seed protocol used here to directly test their capacity-interference hypothesis.
- Increase seed count beyond 5 for both the main comparison and the head-ablation study, to determine whether MSFD’s 0.26-point edge over baseline resolves into significance with more statistical power.
- Extend the KD hyperparameter sweep to a full multi-seed grid rather than a single-seed coarse search.
- Reconcile the Stage-1 progressive-distillation accuracy discrepancy against exact checkpoint provenance.
- Validate on higher-resolution datasets (e.g., a downsampled ImageNet subset) and alternative teacher/student capacity ratios.
- Add an automated train/validation index-overlap assertion as a standard, reusable pre-training check for any future distillation pipeline built on this codebase.

7 Conclusion

We set out to evaluate a multi-headed synthetic-feature distillation method (MSFD) and, in the process of doing so rigorously, found that our own earlier claim of its superiority over standard knowledge distillation did not hold once the comparison was made fair: matched training budgets, a validation-tuned KD baseline instead of an arbitrary one, and five seeds per method with paired significance testing. Under those conditions, a properly tuned standard-KD baseline significantly outperforms both MSFD and a no-distillation control, and MSFD itself does not significantly outperform training the student with no distillation at all. We report this outcome, together with the infrastructure and a documented data-leakage bug that shaped how we got here, as we believe an honestly reported negative result—and the evaluation discipline required to reach it—is more useful to the field than the originally intended positive claim would have been.

8 Recommendation

- For practitioners evaluating a new distillation method: tune the KD baseline’s temperature and loss weighting via a validation-only search before concluding a more complex method is worthwhile. In this study, hyperparameter tuning alone recovered more accuracy than the architectural innovation under test.
- Always measure a no-distillation baseline. Its absence in the original version of this project meant an entire category of possible explanation (“maybe nothing helps here”) was invisible until it was explicitly added.
- Use multiple seeds (5 at minimum, more if compute allows) with paired significance testing for any headline distillation comparison; single-run comparisons in this domain are demonstrably prone to reversing under matched conditions, as shown directly in Section 5.1.

- Add an explicit train/validation split-integrity assertion (disjoint indices, correct sizes) to any custom data pipeline before trusting validation-based model selection; the bug in Section 5.1 produced no crash or warning and was caught only by an anomalous accuracy gap.
- Before adopting a method name, search for prior use of that exact name in the current literature—the collision documented in Section 3.3 was avoidable and cost a full manuscript revision cycle.

Declarations

Author Contributions

Yonas Zewdie: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualization.

Funding

This research received no external funding.

Conflicts of Interest

The author declares no conflicts of interest.

Ethics Statement

This work uses only the publicly available CIFAR-100 dataset and does not involve human subjects, animal subjects, or personally identifiable data. No ethics board review was required.

Data and Code Availability

The training pipeline (teacher training, feature caching, MSFD distillation, KD baseline with hyperparameter sweep, deliberate practice, progressive distillation, and all evaluation/statistics scripts), the trained checkpoints, and the raw experiment logs referenced in this paper are archived with a citable DOI (Zenodo) and the corresponding source code is hosted on a public repository (GitHub) linked from the Zenodo record. All figures in this paper were generated programmatically from logged experiment output, with no manually transcribed or estimated values.

Environment

Component	Value
Hardware	1× NVIDIA Tesla T4 (16 GB VRAM), Kaggle free-tier notebook
Framework	PyTorch with torchvision (CUDA-enabled build), AMP mixed precision
Supporting libraries	scikit-learn (metrics), scipy.stats (significance tests), pandas, tqdm
Dataset	CIFAR-100 (torchvision.datasets.CIFAR100), downloaded at run time

Item	Value
Train / Val / Test sizes	45,000 / 5,000 / 10,000 (verified disjoint by explicit assertion)
Split method	<code>torch.utils.data.random_split</code> , fixed generator seed = 42
Model seeds (main comparison)	42, 139, 236, 333, 430
KD sweep seed	42 (single seed, coarse 6-epoch search)
Batch size / accumulation / effective	32 / 4 / 128
Teacher / student learning rate	1×10^{-4} / 3×10^{-4}
Optimizer / scheduler	AdamW / cosine annealing, 2-epoch linear warmup
Weight decay / gradient clip	1×10^{-4} / 1.0
MixUp alpha	0.2
MSFD distillation / CE weight	0.7 / 0.3
KD temperature / CE weight (tuned)	1.0 / 0.3 (selected via Section 5.2)
Teacher / student / ablation epochs	10 / 12 / 12 (matched)
ViT backbone trainable blocks	Last 4 of 12 (37.9% of teacher parameters)

Data Splits, Seeds, and Hyperparameters

A Extended Per-Class Results

Table 8 extends Figure 8 to the top and bottom 10 classes by test-set F1 score (single-seed evaluation; the full 100-class report is included in the archived code/data repository).

Table 8. Top/bottom 10 classes by F1 score (extending Figure 8).

Rank	Best-performing class	F1	Rank	Worst-performing class	F1
1	sunflower	0.950	1	seal	0.526
2	wardrobe	0.940	2	boy	0.537
3	apple	0.931	3	shrew	0.551
4	skunk	0.931	4	otter	0.580
5	tractor	0.929	5	girl	0.581
6	lawn_mower	0.916	6	man	0.604
7	bicycle	0.905	7	mouse	0.610
8	skyscraper	0.899	8	woman	0.615
9	road	0.899	9	willow_tree	0.616
10	motorcycle	0.891	10	bear	0.618

B Full Ablation Detail

Table 9. Head ablation detail (single seed, 12-epoch matched budget; see Section 5.14.1 for statistical caveats).

Configuration	Test Acc.	Δ vs. full (3-head)	Parameters
visual_only	77.73%	−0.24 pp	1.49M
audio_only	77.40%	−0.57 pp	1.49M
3d_only	77.56%	−0.41 pp	1.49M
visual_audio	77.69%	−0.28 pp	1.89M
visual_3d	77.64%	−0.33 pp	1.89M
full (3 heads)	77.97%	–	2.28M